

(a) Prompt Outline for IntentTrace's Element-Level Diagnostic Generation

You audit a process model against the source text it was built from. Extract from the description the verbatim spans that justify each element, then sort every element and every described step into match, mutation, omission, or addition. {Definition of element types under audit} {Span-extraction rules} {Model element and span matching rules} {Output categories description} Examples: {Few-shot examples} Natural-language Process Specification: {Process Specification} Process Model: {Model under evaluation} {Output format definition}
--

(b) Prompt Outline for IntentTrace's Element-Level (Variable and Actors) Diagnostic Generation

You audit the variable and actor declarations of a generated process model against the source text, sorting every record into match, mutation, omission, or addition. {Definition of element types under audit} {Variable matching rules} {Actor matching rules} {Output categories description} Examples: {Few-shot examples} Natural-language Process Specification: {Process specification} Variable Declarations for Variable Matching: {Variable declarations} Actor Declarations for Actor Matching: {Actor declarations} Node List for Actor Assignment: {Node List} {Output format Definition}
--

(c) Prompt Outline for IntentTrace's Specification Traces Extraction

You read a process and list its distinct execution paths. Each path is the ordered sequence of step IDs from start to finish; together the paths must cover every route, with none invented and none left out. {Trace-construction rules} Examples: {Few-shot examples} Natural-language Process Specification: {Process specification} Specification Spans: {Spec spans} {Output format Definition}

(d) Prompt Outline for Process Model Refinement

You are an expert process model refiner. Your only task is to edit the DSL process model below so it resolves the given feedback and better matches the process description. Grammar of the DSL: {DSL Grammar} Structural, Logical and Specification-Alignment Constraints: {Structural, Logical and Specification-Alignment Constraints} {Refinement rules} Natural-language Process Specification: {Process Specification} Specification Spans: {Spec spans} Specification Traces: {Spec traces} Process Model under Refinement: {Model under refinement} Model Execution Traces: {Model traces} Diagnostic: {Diagnostic} {Output format Definition}

(e) Prompt Outline for Direct-LLM's Diagnostic Generation

You review behavioural process models. Report, precisely where the model does not faithfully reflect the process specification based on the provided specification-alignment constraints. Grammar of the DSL: {DSL Grammar} Specification-Alignment Constraints: {Specification-Alignment Constraints} Natural-language Process Specification: {Process Specification} Process Model: {Model under evaluation} {Output format Definition}
